The two-point orbit in a single-crossing argument

Human Compression project: L044 working continuation

Let $G = \langle \rho_0, \ldots, \rho_{r-1} \rangle \leq A_n$ be a faithful transitive string C-group with every maximal omission subgroup $G_i = \langle \rho_j : j \neq i \rangle$ intransitive. Suppose that precisely one i-edge ab joins distinct G_i-orbits. The accepted single-crossing lemma gives exactly two such orbits A, B, with $a \in A, b \in B$; all generators except possibly $\rho_{i-1}, \rho_i, \rho_{i+1}$ fix a, b. The restrictions of the generators of G_i to either orbit are involutory string generators, but their projected marking need not inherit the intersection property.

Lemma 1 (Primitive orbit, with the small projected exception)**.** *If the action of G_i on A is primitive and $|A| \geq 3$, its nontrivial distinguished restrictions occupy an interval of indices. If $|A| = 2$ and they do not occupy an interval, their indices are exactly $i-1, i+1$, and both restrictions are the same transposition.*

Proof. If the active index set is disconnected, partition it at a gap into two nonempty sets. Their images H, K commute and are nontrivial normal subgroups of the primitive image HK. Each is transitive. Every nonidentity element of one that centralizes the other is fixed-point-free: fixing one point and commuting with a transitive group forces fixation of all points. In particular every active distinguished restriction moves a. The single-crossing lemma therefore confines the entire active set to $\{i-1, i+1\}$. Its two commuting involutions each generate a transitive group, forcing the degree to be two. The degree-two conclusion follows immediately. This argument uses normality and commutation, and does not identify the projected product with a direct product. □

Lemma 2 (An end-adjacent two-point orbit)**.** *Suppose $n \geq 12$, $i = 1$, $A = \{a, a'\}$, and both ρ_0 and ρ_2 interchange a, a'. Then*

$$r \leq \lfloor (n-1)/2 \rfloor.$$

The dual assertion holds for $i = r - 2$.

Proof. Ranks at most three already satisfy the result, so assume $r \geq 4$. Set $z = \rho_0$ and $K = \langle \rho_2, \ldots, \rho_{r-1} \rangle$. The string and intersection properties give $G_1 = \langle z \rangle \times K$. Restriction of G_1 to B is faithful: an element in its kernel is supported on A, and the only possible nonidentity element there is an odd transposition. The restriction z_B is nontrivial, central in the transitive group $(G_1)_B$, and is therefore fixed-point-free. Write $|B| = 2m$. Since z is even and also transposes A, m is odd. The assumption $n \geq 12$ implies $m \geq 5$.

The m pairs of z_B form a block system. The group K is transitive on these blocks because G_1 is transitive on B and z fixes each block. Choose a minimal subfamily L of the displayed K generators whose block images generate the block action. The subgroup $\langle z, L \rangle$ is transitive on B: it is transitive on the pairs and contains their simultaneous flip. If a displayed K generator ρ_k were outside L, then G_k would contain z, L, and ρ_1. These act transitively on B, join B to a by the 1-edge, and join a to a' by z. Hence G_k would be transitive on all n points, a contradiction. Consequently every K generator belongs to L, and their block images form an independent involutory string family of length $q = r - 2$ in S_m. The retained Whiston bound gives $q \leq m - 1$.

If $q \leq m - 2$, the desired bound $r \leq m$ follows. Suppose $q = m - 1$. The retained extremal theorem of the complete L047 common theory gives block image S_m, with an abstract

Coxeter-path marking. For $m \geq 7$ this is the proved common extremal recognition theorem; its separately accepted complete degree-five certificate supplies $m = 5$. In particular all of these block generators are conjugate.

The kernel N of the block action of K lies in the elementary abelian group $V = \mathbb{F}_2^m$ of independent pair flips. It is an S_m-invariant subspace. For completeness its possible subspaces are 0, the constant line $\langle\lambda\rangle$, the even-weight space E, and V. Indeed a nonconstant vector has a transposition translate whose difference has weight two, and the S_m-orbit of that difference spans E. The remaining constant vectors give the stated list. Since m is odd, $\lambda \notin E$.

The constant flip λ is exactly z_B. It cannot belong to N: faithfulness of the restriction would put z in K, contrary to IP. Therefore $N = 0$ or $N = E$. Both consist of even permutations of B, so sgn_B on K descends to a homomorphism from its block image S_m to C_2. Such a homomorphism has the same value on conjugate block generators. This contradicts the actual signs: ρ_2 transposes A and is even globally, so is odd on B, whereas $\rho_3, \ldots, \rho_{r-1}$ fix A and are even on B. There is at least one of the latter generators, since $q = m - 1 \geq 4$. Thus equality $q = m - 1$ is impossible, finishing the proof. Reversing all indices proves the dual statement. □

Lemma 3 (The remaining internal case has two intransitive factors)**.** *Suppose $|A| = 2$, its two active indices are $i-1, i+1$, and $2 \leq i \leq r-3$. Put $H = \langle\rho_0, \ldots, \rho_{i-1}\rangle$ and $K = \langle\rho_{i+1}, \ldots, \rho_{r-1}\rangle$. Neither restricted factor is transitive on B.*

Proof. Again restriction of $G_i = H \times K$ to B is faithful. If H_B were transitive, every nonidentity element of K_B would be fixed-point-free, since it centralizes H_B. All distinguished generators in K are nonidentity on B by faithfulness. Its involutions would therefore all have the same sign $(-1)^{|B|/2}$ there. But ρ_{i+1} must be odd on B (it swaps A), while ρ_{i+2} is even on B (it fixes A). This is impossible. Interchanging the factors gives the other assertion. □

Lemma 4 (Odd commuting factors on a transitive set)**.** *Let $H, K \leq S_B$ commute, with HK transitive on B of size $b \geq 10$. Suppose each factor is intransitive, each contains an odd permutation of B, and each is a string C-group with its displayed faithful marking, of ranks $p, q \geq 2$. Then*

$$p + q \leq \lfloor (b-1)/2 \rfloor.$$

The retained input is the complete transitive string C-group bound of L047, not the alternating-group theorem being recovered here.

Proof. The group K is transitive on the set of H-orbits, because HK is transitive. Moreover it transports their H-actions identically: it commutes with every element of H. Thus restriction of H to one of its orbits is faithful. Indeed an element trivial on that orbit is trivial on every K-translate, hence on B. In particular this restriction preserves the full distinguished marking and IP. All H-orbits have a common size h, say there are t of them, and every element of H has the same sign on each orbit. An odd element of H therefore forces t to be odd. Since H is intransitive, $t \geq 3$; hence $h \leq b/3$. Whiston's bound and $p \geq 2$ give $h \geq 3$. Reversing the roles gives a faithful transitive K-action of degree k with $3 \leq k \leq b/3$, and an odd number of K-orbits. Each of these two faithful actions contains an odd permutation, so neither image is its alternating group.

Suppose first that neither orbit image is the full symmetric group. The accepted L047 bound, including all its small exceptions, gives $p \leq h/2 + 1$ and $q \leq k/2 + 1$. For $b \geq 15$ it follows that

$$p + q + 1 \leq (h + k)/2 + 3 \leq b/3 + 3 \leq (b + 1)/2.$$

If $10 \leq b \leq 14$, the factorization $b = ht$ with $h \geq 3$ and odd $t \geq 3$ forces $b = 12$ and $h = 4$; the same argument gives $k = 4$. Neither faithful orbit image is S_4, so Whiston's equality clause gives $p, q \leq 2$, which is stronger than required.

It remains to treat, up to interchange, an orbit image $H = S_h$. The centralizer of the natural S_h on an H-orbit is trivial, since $h \geq 3$. Identify each H-orbit with a fixed one by an element of K transporting the two orbits. This identification is independent of the chosen transporter: their difference centralizes H and stabilizes that orbit, hence fixes it pointwise. Thus a commuting permutation is determined entirely by how it permutes the t copies: there are no nontrivial internal coordinate maps. It follows that K has a faithful transitive action on the t copies, and each of its orbits on B has size t. Whiston now gives $p \leq h - 1$ and $q \leq t - 1$. Since $h, t \geq 3$,

$$p + q + 1 \leq h + t - 1 \leq (ht + 1)/2 = (b + 1)/2,$$

where the middle inequality is $(h - 2)(t - 2) \geq 1$. This proves the assertion in every case. □

Lemma 5 (The internal two-point exception is below the target rank)**.** *Under the hypotheses of the preceding internal-case lemma, with $n \geq 12$, one has $r \leq \lfloor (n - 1)/2 \rfloor$.*

Proof. Restriction of $H \times K = G_i$ to B is faithful, as already proved, and it is transitive there. Both factors are intransitive by the preceding lemma. Each contains an odd permutation of B, namely its generator next to the omitted index; global evenness and its transposition of A give precisely that sign. Both retain their full C-markings under the faithful restriction, and have ranks $i \geq 2$ and $r - i - 1 \geq 2$. Apply Lemma 4 with $b = n - 2$ to obtain $r - 1 \leq \lfloor (n - 3)/2 \rfloor$, as required. □

Lemma 6 (The interval reduction in the main contradiction range)**.** *If $n \geq 12$, $r > (n - 1)/2$, and the G_i-action on one of its two orbits is primitive, the nontrivial restrictions to that orbit occupy an interval of indices.*

Proof. For orbit size at least three this is the first lemma. A singleton has empty active set. In degree two the only noninterval possibility has both neighbors of i active. An end-adjacent i is excluded by the second lemma, and all other internal indices are excluded by the preceding lemma. No two-sided exception exists at an endpoint. □

Attribution, dependencies and remaining obligation. The primitive-orbit interval assertion is source Proposition 5.1, physical page 22 of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1. Its projected direct-product claim is not used here. The degree-two possibility is a boundary left by that proof, not a claimed counterexample satisfying the complete source hypotheses. The end-adjacent exclusion above is an additional argument using the entire accepted L047 common theory and its inherited extremal classification, including its degree-five finite premise. These inputs must be paid in any complete application. The odd commuting factor lemma and the final application close the internal case using the same accepted transitive rank theorem at a smaller faithful degree. No new finite search is needed and the desired alternating-group theorem is not an input. Thus the two-point exception left by the elementary primitive-orbit argument does not affect the main rank bound. This does not prove the source's literal interval statement in that exceptional case, or classify its small markings. Neither a whole-source recovery nor a new conjecture result or formal proof is asserted by this working note.